

Project Overview: Real-Time Data Processing and Analytics for an E-commerce Platform

Objective:

Participants will develop a real-time data processing system for an e-commerce platform that collects and processes data such as customer interactions, orders, product views, and sales in real-time. The project will involve creating a data pipeline that collects real-time data from a streaming source (like Kafka or a simulated stream), transforms the data, and loads it into a data warehouse. The data will be processed and analyzed using Spark, Databricks, and other cloud services like Azure.

Week 1: Data Warehousing and SQL Concepts

Topics Covered:

- Introduction to DWH concepts and importance in real-time analytics
- SQL basics for creating, querying, and managing data in the warehouse
- Using SQL DML statements for data manipulation and query optimizations

Capstone Project Milestone:

- **Objective:** Design a Data Warehouse schema to store real-time e-commerce data (e.g., customer orders, product interactions).

Tasks:

1. **Design Schema:** Create tables to store customer data, product data, and order data.
2. **Data Manipulation:** Insert, update, and manage data using SQL.
3. **Querying Data:** Write SQL queries to analyze customer behavior, order data, and product interactions.

Example Code:

```
-- Create schema for e-commerce platform
CREATE TABLE customer_dim (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(255),
    location VARCHAR(255)
);

CREATE TABLE product_dim (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(255),
    category VARCHAR(255)
);

CREATE TABLE order_fact (
    order_id INT PRIMARY KEY,
    product_id INT,
    customer_id INT,
    quantity INT,
    order_amount DECIMAL(10, 2),
    order_date TIMESTAMP
```

```
);

-- Insert sample data into the tables
INSERT INTO product_dim VALUES (1, 'Smartphone', 'Electronics');
INSERT INTO customer_dim VALUES (2001, 'Bob', 'California');
```

Outcome: By the end of Week 1, participants will have designed a Data Warehouse schema and managed data with SQL for real-time e-commerce data.

Week 2: Real-Time Data Processing with Python

Topics Covered:

- Python fundamentals for data extraction, transformation, and loading
- Real-time data collection using simulated streams or Kafka
- Python-based ETL processing for real-time data

Capstone Project Milestone:

- **Objective:** Use Python to extract and process real-time e-commerce data and load it into the Data Warehouse.

Tasks:

1. **Simulate Real-Time Data:** Use Python to simulate real-time data streams (e.g., customer interactions, orders) and load them into the Data Warehouse.
2. **ETL Pipeline:** Develop a real-time ETL pipeline to transform and load the data into the warehouse.

Example Code:

```
import pandas as pd
from sqlalchemy import create_engine
import random
from datetime import datetime

# Simulate a real-time data stream for orders
def simulate_order_stream():
    product_ids = [1, 2, 3, 4]
    customer_ids = [2001, 2002, 2003]

    while True:
        order = {
            'order_id': random.randint(1001, 9999),
            'product_id': random.choice(product_ids),
            'customer_id': random.choice(customer_ids),
            'quantity': random.randint(1, 5),
            'order_amount': random.uniform(50, 500),
            'order_date': datetime.now()
        }
        yield pd.DataFrame([order])

# Stream and load real-time data into a SQL table
engine = create_engine('sqlite:///ecommerce.db')

for order_batch in simulate_order_stream():
```

```
order_batch.to_sql('order_fact', con=engine, if_exists='append', index=False)
print("Order batch loaded into the warehouse.")
```

Outcome: By the end of Week 2, participants will have implemented a real-time ETL pipeline using Python to simulate and process streaming e-commerce data.

Week 3: Real-Time Data Processing with Apache Spark and PySpark

Topics Covered:

- Introduction to Apache Spark for real-time processing
- PySpark for handling large-scale real-time data

Capstone Project Milestone:

- **Objective:** Process and analyze real-time data using Apache Spark and PySpark.

Tasks:

1. **Set Up Spark Streaming:** Configure Apache Spark and PySpark to process real-time data streams.
2. **Data Processing with PySpark:** Use PySpark to process large-scale, real-time e-commerce data and analyze it in real-time.

Example Code:

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import col

# Create Spark session
spark = SparkSession.builder.appName("EcommerceRealTime").getOrCreate()

# Read data from the streaming source
order_df = spark.readStream.format("csv").option("header",
"true").load("path_to_streaming_data")

# Perform real-time analysis using PySpark
product_sales = order_df.groupBy("product_id").sum("order_amount")

# Write the results to a console (or database)
query = product_sales.writeStream.outputMode("complete").format("console").start()

query.awaitTermination()
```

Outcome: By the end of Week 3, participants will be able to handle real-time data processing and analytics using Apache Spark and PySpark.

Week 4: ETL Pipelines in Azure Databricks

Topics Covered:

- Azure Databricks features for real-time data engineering
- Delta Live Tables and ETL Pipelines
- Azure Data Factory for workflow automation

Capstone Project Milestone:

- **Objective:** Build real-time ETL pipelines using Azure Databricks and Delta Live Tables.

Tasks:

1. **Configure Databricks:** Set up Azure Databricks and build real-time ETL pipelines for streaming data.
2. **Process Data with Delta Live Tables:** Automate data processing and incremental updates with Delta Live Tables in Databricks.

Example Code:

```
# Create Delta Live Table for real-time data processing
CREATE OR REPLACE LIVE TABLE orders_delta
AS SELECT * FROM streaming.`/mnt/ecommerce/orders/stream`;

# Define the ETL pipeline in Databricks using Delta Live Tables
CREATE OR REPLACE STREAMING LIVE TABLE enriched_orders
AS SELECT order_id, product_id, customer_id, order_amount
FROM orders_delta;
```

Outcome: By the end of Week 4, participants will have built and deployed real-time ETL pipelines in Azure Databricks.

Week 5: Deploying the Real-Time Pipeline with Azure DevOps

Topics Covered:

- Azure DevOps for pipeline automation and deployment
- CI/CD for deploying real-time data processing pipelines

Capstone Project Milestone:

- **Objective:** Deploy the real-time data pipeline using Azure DevOps for automation and monitoring.

Tasks:

1. **Azure DevOps Pipelines:** Set up Azure DevOps pipelines for continuous integration and deployment of the real-time data pipeline.
2. **Automate Deployment:** Automate the deployment of real-time data processing jobs using Azure DevOps.

Example Code:

```
# Azure DevOps pipeline YAML configuration for real-time pipeline
trigger:
  - main

pool:
  vmImage: 'ubuntu-latest'

steps:
  - task: UsePythonVersion@0
    inputs:
      versionSpec: '3.x'
```

```
- script: |  
    pip install -r requirements.txt  
    python real_time_pipeline.py  
    displayName: 'Run Real-Time Pipeline'
```

Outcome: By the end of Week 5, participants will have automated the deployment of real-time pipelines using Azure DevOps.

Summary of Outcomes:

1. **Week 1:** Design a Data Warehouse schema and perform SQL operations for real-time data.
2. **Week 2:** Build a Python-based real-time ETL pipeline for processing and storing data.
3. **Week 3:** Use Apache Spark and PySpark to process large-scale real-time data.
4. **Week 4:** Implement real-time ETL pipelines with Azure Databricks and Delta Live Tables.
5. **Week 5:** Deploy and automate the real-time data pipelines using Azure DevOps.